MACHINE LEARNING FOR NOISY POWER SYSTEM STATE ESTIMATION

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ABSTRACT

The increasing electrification of the world demands efficient monitoring of power distribution systems to manage distributed energy resources and ensure grid stability. Distribution grids have poor observability because it is too expensive to install sensors at each node, whereas transmission networks are closely watched. Consequently, operators often rely on sparse, low-fidelity (LOFI) measurements, hindering the precise estimation of bus voltages and phase angles required for optimal control. However, current physics-based models struggle to scale effectively or handle observational noise when reconstructing high-fidelity (HIFI) states from these limited inputs. Here we present a Physics-Informed Graph Neural Network (PIGNN) framework that trains on voltage data derived from LOFI power measurements via traditional solvers to predict HIFI voltage states using interpolated power inputs. By integrating Kirchhoff's laws and topology constraints into the learning process, our approach is expected to maintain robustness against observational noise while accurately inferring HIFI voltage profiles that purely physics-based solvers cannot directly observe. Validated against the realistic SMART-DS synthetic datasets, this methodology offers a scalable solution for real-time state estimation in data-scarce environments, facilitating the reliable integration of renewable energy sources into modern smart grids.

1 MOTIVATION

The world is becoming increasingly electrified and, at the same time, trying to limit the emissions caused by this increased need for power generation. One way to address this is to improve the efficiency of how electricity is distributed from its source to its destination. This can be done through state estimation of the bus systems of distribution systems, and more specifically, by accurately estimating the bus voltage and bus voltage phase angles of these buses. The electrical grid can be broken up into two portions: a transmission grid and a distribution grid. The transmission grid contains the larger power lines that connect the high-voltage power stations to the rest of the grid. The distribution grid, on the other hand, manages the lower voltage lines and distributes the electricity from substations to its final destination, such as homes or office buildings.

While the transmission grid is heavily measured and its efficiency is highly optimized, the distribution grid is often only measured in a few spaces because placing measurement devices on every power line would be too costly for the utility company. As a result, utilities try to utilize the few measurements they do have and then estimate the usage rates of the distribution grid's power lines in order to distribute electricity in the most efficient way possible.

2 HOW IT CAN RELATE TO A DATA PROBLEM OR A MODELING PROBLEM

Traditionally, physics-based models are the norm when it comes to estimating these missing measurements, but in recent years, research has been done into integrating machine learning techniques to improve these models. Physics-based models often fall short due to the lack of data in distribution systems, and machine learning models can help make more accurate estimates of usage, leading to

improved efficiency. The sparse distribution grid datasets can lead purely physics-based models to over-simplify their predictions, and purely data-driven methods would likely result in outputs that are not physically feasible. To address this, researchers have recently begun investigating methods to incorporate scientific machine learning to fill in the gaps in measurement to find ways to improve grid efficiency.

Although previous research has worked on implementing PINN models to improve efficiency, current methods fall short when it comes to the scalability of the models and their robustness to noise (Ngo et al.). We will be primarily including Gaussian white noise, as well as introducing some non-Gaussian methods such as dropout. Following the work done by Vanin et al., we chose to use a 1% to 5% noise interference error level, which is above typical real-world scenarios, but commonly used on synthetic data. We will be utilizing Graph Neural Networks (GNNs), Physics Informed Neural Networks (PINNs), and Physics Informed Graph Neural Networks (PIGNNs) as our base model and using the results of Ngo et al. as a baseline. PIGNNs are also an attractive approach to the problem because they have the flexibility of GNNs and can adapt to changes in topology, as the physical constraints enable the model to adapt to topology and grid size changes.

3 Prior work

3.1 Relevant Work

State estimation in power distribution networks has traditionally relied on physics-based approaches such as Weighted Least Squares (WLS) estimation and iterative power flow solvers. These methods perform well in transmission systems with dense measurement coverage, but they struggle in distribution networks where sensor placement is sparse and noisy (Abur & Exposito, 2004). To address these limitations, recent work has explored hybrid scientific machine learning methods that incorporate physical priors into data-driven models.

Physics-informed approaches have recently been applied to power system learning tasks. For example, Tuo et al. train a Graph Neural Network to predict power-flow outcomes in transmission networks by leveraging graph structure and historical injections (Tuo et al., 2023). Meanwhile, Pan et al. propose the DeepOPF framework—a deep neural network approach to accelerate AC-OPF solutions while explicitly enforcing equality and inequality constraints via a penalty scheme (Pan et al., 2020). These works demonstrate the value of combining learning with physical constraints; however, they either target transmission networks, optimal power flow problems (rather than state estimation), or assume ideal measurement conditions.

More recently, Physics-Informed Graph Neural Networks (PIGNNs) have emerged as a promising hybrid solution. Ngo et al. demonstrated that combining GNN message passing with embedded power flow constraints improves estimation accuracy over both PINNs and standard GNNs, particularly in low-observability settings (Ngo et al., 2024). Their approach, however, assumes noise-free training labels and relies on custom datasets, limiting real-world applicability. Our work extends this direction by incorporating realistic measurement noise, evaluating performance on the widely-used SMART-DS dataset, and explicitly studying the impact of noise and solver bias in low-fidelity data.

3.2 Power System Data Generation

Low-Fidelity (Lo-Fi) data, which usually consists of active (P) and reactive (Q) power measurements recorded at coarse temporal intervals (e.g., every 15 minutes), frequently limits operators in practical distribution network monitoring. Conventional Power Flow Solvers (like OpenDSS) are computationally costly and iterative, making them inappropriate for real-time, high-frequency state estimation even though they can compute the corresponding system states (Voltage Magnitude and Phase Angle) with high precision.

As illustrated in the project's conceptual workflow (Figure 1), the data generation process transforms low-fidelity active (P) and reactive (Q) power inputs into a rich, high-fidelity dataset through two key mechanisms:

1. **Training Phase (Lo-Fi Domain):** We utilize the available Lo-Fi power data and pass it through a traditional solver to generate accurate ground-truth voltage labels. These pairs of

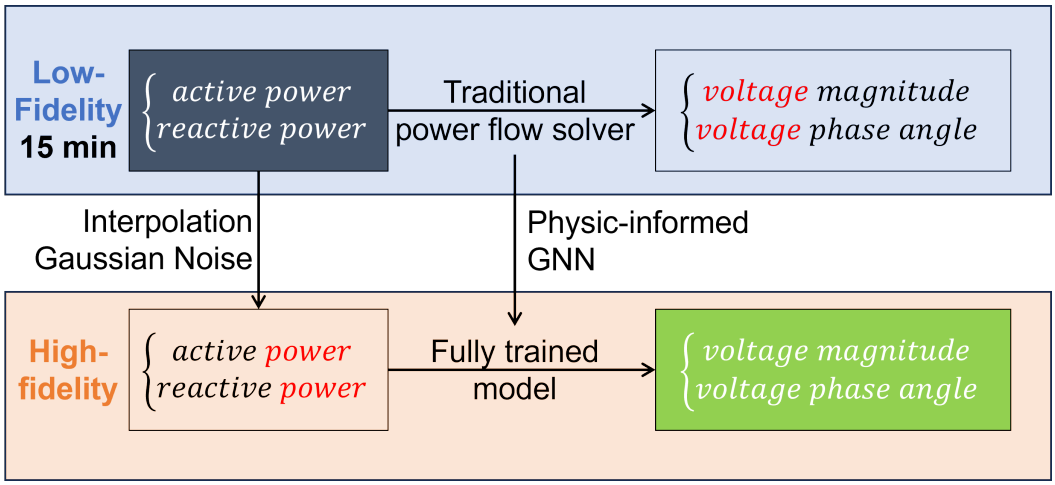

Figure 1: Conceptual workflow

$(P, Q)_{LoFi} \rightarrow (V, \theta)_{LoFi}$ form the training dataset for our Physics-Informed Graph Neural Network (GNN).

2. **Inference Phase (Hi-Fi Domain):** Once trained, the ML model enables high-fidelity analysis. We obtain High-Fidelity (Hi-Fi) power inputs via mathematical interpolation of the coarse data. The ML model then predicts the corresponding Hi-Fi voltage states almost instantaneously, bypassing the slow traditional solver.

3.2.1 System Architecture and Topology Construction

Circuit Model Initialization We simulate the `p4uhs0_4--p4udt4` distribution feeder using OpenDSS. The setup validates essential circuit files and load data before initializing the model, which consists of:

- **71 Bus Nodes:** The structural connection points of the grid.
- **36 Load Objects:** Individual points of power consumption.

Heterogeneous Graph Construction A key step is translating the electrical circuit into a heterogeneous graph structure (G). This graph explicitly distinguishes between two types of nodes to better model the grid's physics:

- **Bus Nodes (Type 0):** Represent the grid's skeleton. Their features are initialized to zero, and their target labels are the system states (Voltage Magnitude and Angle).
- **Load Nodes (Type 1):** Represent power consumers. Their features are dynamic power injections (kW, $kVar$), while their voltage states are treated as latent.

The resulting topology, visualized in Figure 2, contains 107 nodes connected by power lines and load connections. The layout emphasizes the radial structure of the distribution network, with loads positioned around their parent buses.

3.2.2 Temporal Data Integration and Simulation

Dynamic Load Mapping To simulate realistic conditions, we map specific OpenDSS load objects to external time-series data profiles (e.g., `com_16145.parquet`). This ensures that the simulation reflects non-uniform demand patterns across the grid rather than static averages.

Sample Generation Pipeline We generate training samples by iterating through specific timepoints. The process involves:

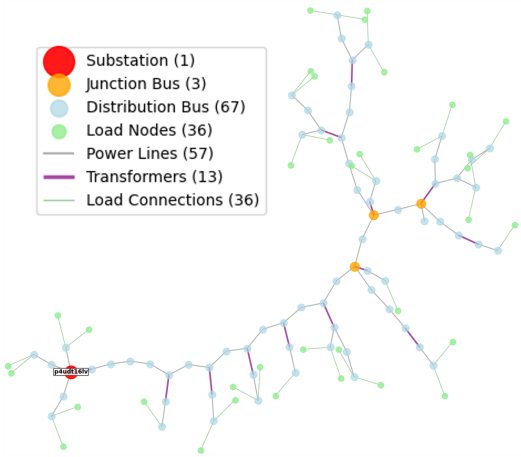

Figure 2: Disaggregated Power Network Topology.

1. **Data Extraction:** Retrieving active and reactive power values for each load.
2. **Noise Injection:** Adding Gaussian noise to load profiles to simulate measurement uncertainty and improve model robustness.
3. **Solution & Extraction:** Solving the power flow with OpenDSS and extracting input features (P, Q) and output labels (V, θ).

A total of 50 samples were generated for this dataset.

3.2.3 Dataset Analysis

The produced dataset was examined to guarantee quality and transformed into a format suitable for machine learning. The statistical distribution of important variables is shown in Figure 3. The histograms confirm the heterogeneous nature of the data: While bus nodes (blue) display voltage profiles clustered within typical operating limits, load nodes (green) display a wide range of power consumption. The non-linear relationship between load power and bus voltage, which the ML model seeks to understand, is depicted in the correlation plot (bottom right).

3.2.4 Deliverables

The final processed data is saved as `disaggregated_power_system_dataset.npz`. This file contains the complete topology, feature matrices, and label matrices, serving as the foundation for training our Graph Neural Network.

4 What will the baseline be

Because the usage of scientific machine learning in distribution state estimation is a relatively new topic, there are few comparable baselines for us to utilize. However, we plan to use the PIGNN model put forth by Ngo et al. as the barometer to test the success of our model. Ngo et al.'s model would be a good baseline for us because their model achieved good results on the datasets they used, but they also mentioned that they did not incorporate noise when training their model. We hope to achieve similar results to their model, while being more robust to noise and remaining accurate when scaled to real-world grid sizes.

However, the model by Ngo et al. is not completely comparable to our model because they utilized their own dataset, and we will be utilizing the SMART-DS dataset. As a result, we implemented a baseline GNN model that uses the same SMART-DS dataset when training, so we can have a more direct comparison during the model development process.

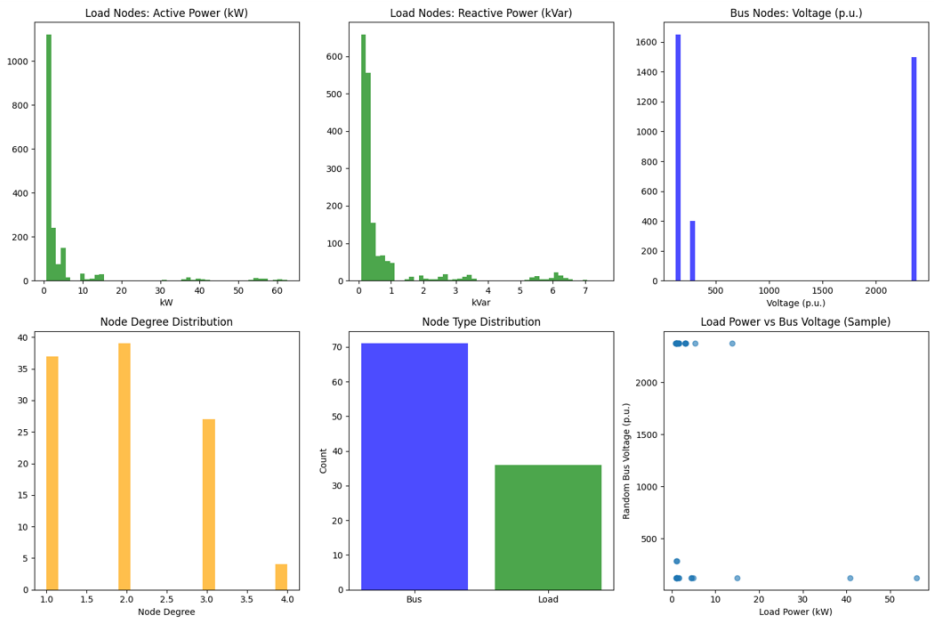

Figure 3: Statistical Analysis of the Disaggregated Dataset.

5 Successful Results

To demonstrate the effectiveness of our methodology, we expect several key outcomes that would indicate successful high-fidelity state estimation under noisy and low-observability conditions. First, we anticipate that the Physics-Informed Graph Neural Network (PIGNN) will outperform both the baseline GNN and physics-only solvers across voltage magnitude and phase angle estimation tasks. A successful result would show a consistent reduction in mean absolute error and root-mean-square error, ideally in the range of 20–40% improvement over the baseline GNN when evaluated on the SMART-DS feeder.

Moreover, we expect the PIGNN to maintain robustness as noise levels increase. Under Gaussian noise between 1–5%, the model should produce stable voltage reconstructions without large deviations at weakly observed nodes or feeder endpoints. Performance should degrade gracefully at higher noise levels, demonstrating the model's ability to generalize beyond the distribution of training perturbations.

An additional indicator of success would be the model's adherence to physical constraints. We anticipate that Kirchhoff's Current Law (KCL) residuals will remain small during inference, showing that the physics-informed loss meaningfully guides the model toward physically consistent solutions. Low residuals across the entire distribution feeder would confirm that the model has internalized structural grid relationships beyond what a data-driven GNN can capture.

Finally, we expect qualitative improvements in voltage profile smoothness and spatial coherence. Successful results would include voltage trajectories that follow realistic power flow behavior, capturing nonlinear sensitivity to localized load variations and the propagation of these effects across the network. If these expectations are met, the results would validate that combining graph-based architectures with physics-informed regularization provides a scalable, noise-resilient solution for distribution system state estimation.

6 Risks related to the methodology

This research employs a methodology where Machine Learning (ML) models, including Physics-Informed Graph Neural Networks (PIGNNs), are trained on Low-Fidelity (LOFI) power data pro-

cessed by traditional solvers, and subsequently evaluated on interpolated High-Fidelity (HIFI) data. The following risks are associated with this approach:

- **Interpolation Bias:** Deriving HIFI power inputs via interpolation inherently smooths out high-frequency stochastic fluctuations. When compared to the model's performance on real, volatile high-frequency sensor data, evaluation on this dampened data may result in an overestimation of the model's accuracy.
- **Solver Dependence:** Training labels are generated by physics-based solvers (e.g., OpenDSS) rather than empirical measurements. The ML model essentially learns to approximate the solver's mathematical abstractions, inheriting any assumptions or parameter inaccuracies present in the synthetic SMART-DS dataset.
- **Physics-Informed Optimization Stability:** Effectively balancing the data-driven loss against the physics-based loss (Kirchhoff's laws) is challenging. Improper hyperparameter tuning risks producing a model that either fits training noise or fails to strictly satisfy physical constraints during inference.
- **Noise Distribution Mismatch:** Real-world conditions are made simpler by using synthetic Gaussian white noise (1%–5%). The model might not be resilient to sensor dropouts, non-Gaussian measurement errors, or outliers that are common in real distribution systems.

7 Potential Mitigation of Risks

Mitigations. Several strategies can reduce the risks associated with training on LOFI solver data and evaluating on interpolated HIFI inputs:

- **Interpolation Bias:** Use multi-fidelity training with mixed LOFI/HIFI data; inject realistic high-frequency components into interpolated traces; evaluate across multiple stochastic interpolations; and validate on any available real high-frequency measurements.
- **Solver Dependence:** Apply domain randomization by perturbing solver parameters; train with multi-solver or ensemble labels; calibrate solver outputs using limited real measurements; and treat solver labels as uncertain using robust or probabilistic loss functions.
- **Physics-Informed Optimization Stability:** Use adaptive loss weighting (e.g., uncertainty-based or GradNorm), constrained or Lagrangian formulations, and architectures that enforce some physics by construction. Track physics-violation metrics and employ curriculum strategies that gradually strengthen physical constraints.
- **Noise Distribution Mismatch:** Simulate realistic sensor faults (dropouts, bursts, drift) instead of only Gaussian noise; use robust losses (Huber/Tukey) and outlier-aware preprocessing; incorporate uncertainty estimation; and perform stress tests under diverse noise scenarios.

References

Ali Abur and Antonio Gomez Exposito. *Power System State Estimation: Theory and Implementation*. CRC Press, 2004.

Quang-Ha Ngo, Bang L.H. Nguyen, Tuyen V. Vu, Jianhua Zhang, and Tuan Ngo. Physics-informed graphical neural network for power system state estimation. *Applied Energy*, 358:122602, 2024. ISSN 0306-2619. doi: https://doi.org/10.1016/j.apenergy.2023.122602. URL `https://www.sciencedirect.com/science/article/pii/S0306261923019669`.

Xiang Pan, Minghua Chen, Tianyu Zhao, and Steven H. Low. Deepopf: A feasibility-optimized deep neural network approach for ac optimal power flow problems. *arXiv preprint arXiv:2007.01002*, 2020.

Mingjian Tuo, Xingpeng Li, and Tianxia Zhao. Graph neural network model for power flow prediction. *arXiv preprint arXiv:2307.02049*, 2023.